THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of letters among H, O, X, I, E with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

2. The sum of the angles of rotation of a square and a regular pentagon (in degrees) is: 1

(a) 162 **(b)** 172 **(c)** 152 **(d)** 192

3. A figure has rotational symmetry of order 6 and one of its lines of symmetry. The total number of lines of symmetry it must have is: 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

4. A regular polygon has 8 sides. The number of different angles (less than 360°, other than 0) through which it can be turned to coincide with itself is: 1

(a) 7 **(b)** 9 **(c)** 8 **(d)** 6

5. A regular polygon is rotated again and again through its smallest angle 24°. The number of turns needed to complete a full turn is: 1

(a) 16 **(b)** 17 **(c)** 14 **(d)** 15

6. A regular 9-gon has side 4 cm. Its perimeter (cm) is: 1

(a) 32 **(b)** 40 **(c)** 36 **(d)** 45

7. The image of a point 12 units from a mirror line is how far from the original point (units)? 1

(a) 36 **(b)** 25 **(c)** 24 **(d)** 12

8. A regular polygon has an angle of rotation 72° and its perimeter is 45 cm. The side length is (cm): 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 10

9. **Assertion (A):** The smallest angle of rotation of a regular hexagon is 60°. 1

Reason (R): The smallest angle of rotation is 360° divided by the order of rotational symmetry, and the order is 6.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** An equilateral triangle has rotational symmetry of order 3. 1

Reason (R): An equilateral triangle has 3 lines of symmetry.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A figure has rotational symmetry of order 6. List all the angles (up to but not including 360°) through which it can be turned to coincide with itself. 2

12. A regular polygon has rotational symmetry with the smallest angle of 20° and each side 3 cm. Find the number of sides and its perimeter. 2

13. A figure fits onto itself after turns of 72° and 216° but not after any smaller turn than 72°. Find its order of symmetry and show that 216° is a multiple of 72°. 2

14. A figure has rotational symmetry of order 8. List all the angles (up to but not including 360°) through which it can be turned to coincide with itself. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A designer wants a symmetrical logo with 10 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 10 petals are placed. 3

16. A regular polygon has 12 sides (12 is even). Of its 12 lines of symmetry, how many pass through two opposite vertices and how many pass through the midpoints of two opposite sides? 3

17. A regular 12-gon is rotated about its centre by 210° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 210°. 3

OR

A designer wants a symmetrical logo with 12 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 12 petals are placed. 3

18. A regular polygon has 14 sides (14 is even). Of its 14 lines of symmetry, how many pass through two opposite vertices and how many pass through the midpoints of two opposite sides? 3

OR

A regular 5-gon is rotated about its centre by 288° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 288°. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher shows four shapes: an equilateral triangle, a rectangle (not a square), an isosceles triangle (not equilateral) and a scalene triangle. 5

(i) How many lines of symmetry does the isosceles triangle have?

(ii) How many does the scalene triangle have?

(iii) What is the order of rotational symmetry of the rectangle?

(iv) Which shape has the most lines of symmetry, how many, and how does that compare with the rectangle?

20. For a regular 5-gon of side 8 cm: 5

(a) find the number of lines of symmetry,

(b) the angle of rotation,

(c) the perimeter,

(d) the angle turned by a vertex when the polygon is rotated 4 times through the smallest angle.

OR

Two regular polygons have 3 and 5 sides. Find their smallest angles of rotation, the difference, and the least angle (in degrees) that is a turn for both (the LCM of the two angles). 5

END OF QUESTION PAPER — Total: 40 Marks